Memo: Q4 2024 sales data issue

To: Customer Success | **From:** Data Operations | **Date:** November 5, 2026
Re: Client feedback that "something feels off" with their data

Problem

Over 99% of the client's data is fine. We found one contained glitch (see chart in Appendix C):

- For 3 products (SKU-002, SKU-004, SKU-012) in 25 states, the numbers froze from **Nov 8 to Nov 24, 2024**.
- Each product/state reported the same number every day, at about **half** its normal level.
- That is 1,275 of 230,000 daily records (about 0.5%) and 55,488 units missing.
- The 25 states are exactly the first 25 alphabetically (AK through MS). Real buying doesn't follow the alphabet: this was a processing error on our side, not a change in customer behavior.

Impact

The gap is small (under 1% of Q4 units) but concentrated, so it can steer decisions the wrong way:

- **A slowdown that didn't happen.** These products look like they fell 20% from October to November. The real drop was 6%, the same as every other product.
- **Wrong top sellers.** SKU-002 and SKU-004 are really the #2 and #3 sellers nationally. The data shows them as #4 and #5.
- **Bad planning inputs.** Reorders, forecasts or regional comparisons that use Nov 8–24 undercount these products by about half in those 25 states.

Proposed Solution

- **Now:** A corrected file is ready to send. We replaced the frozen values with each product/state's own daily average from Oct 1 to Nov 7. Tested on the 25 states that weren't affected, it came within 0.4% of actual totals (about ±7% for a single product in a single state), so treat it as an interim estimate.
- **Final:** Re-run those records from the source and redeliver.

What to tell the client: "You were right. A processing error understated three products in 25 states for Nov 8–24. Everything else checks out. A corrected file is attached, and final numbers will follow."

Go Forward Plan

- **Frozen-value check:** before every delivery, flag any product/state that repeats the exact same number for 7+ days (14+ for slow sellers, which can repeat by chance).
- **Peer check:** flag any product/state that drops or spikes for several days versus other states or its own recent average.
- **Compare to last delivery:** reconcile each file against the previous one.
- **Hold if a check fails:** the delivery waits for review, so the client never finds our errors first.
- **Fix the cause:** find and fix the step that wrote half-values for the AK–MS batch, and add a test for it.

Appendix

A. Method

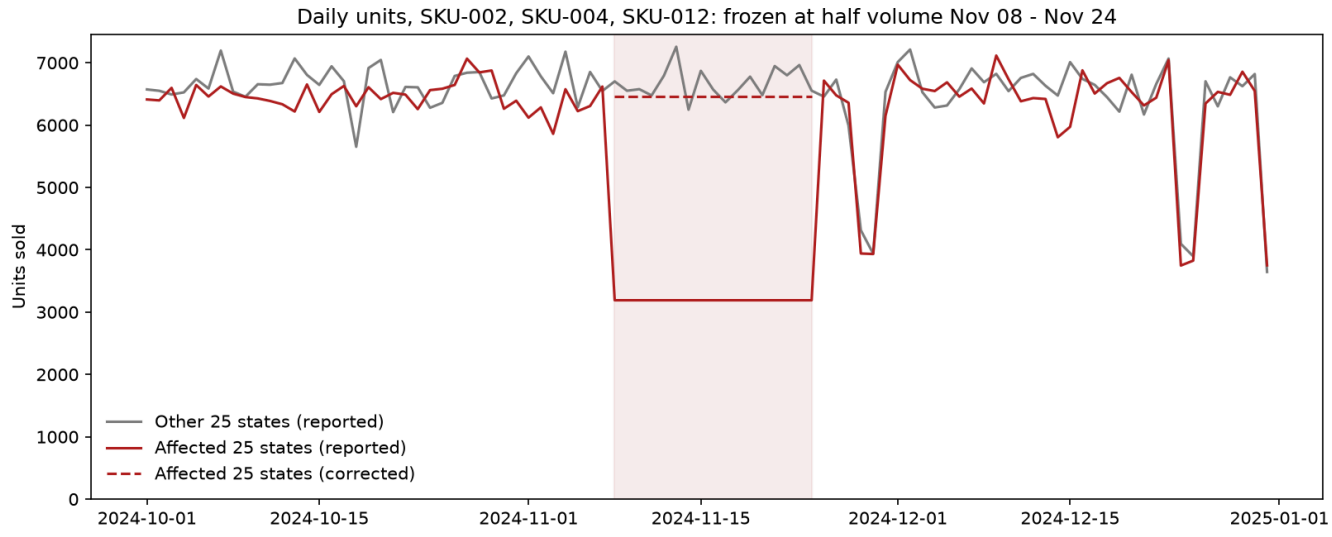
1. Checked the full Q4 2024 file (Oct 1 to Dec 31; 230,000 daily product/state records) for missing days, duplicates, negatives and odd values. None found.
2. Looked for any product/state reporting the exact same number several days in a row. Only one pattern showed up: SKU-002, SKU-004 and SKU-012 in the first 25 states (AK through MS), Nov 8 to Nov 24.
3. Compared those series with the other 25 states and with their own Oct 1 to Nov 7 history. The stuck value equals half the Oct 1 to Nov 7 daily average for each product/state.
4. Replaced the 1,275 stuck values with each product/state's own Oct 1 to Nov 7 daily average. To check this, we applied the same method to the same window in the 25 unaffected states: totals came within 0.4% of actual, and about ±7% for a single product/state.
5. Ruled out a few patterns that looked wrong at first but are normal (see D).

B. Impact by product

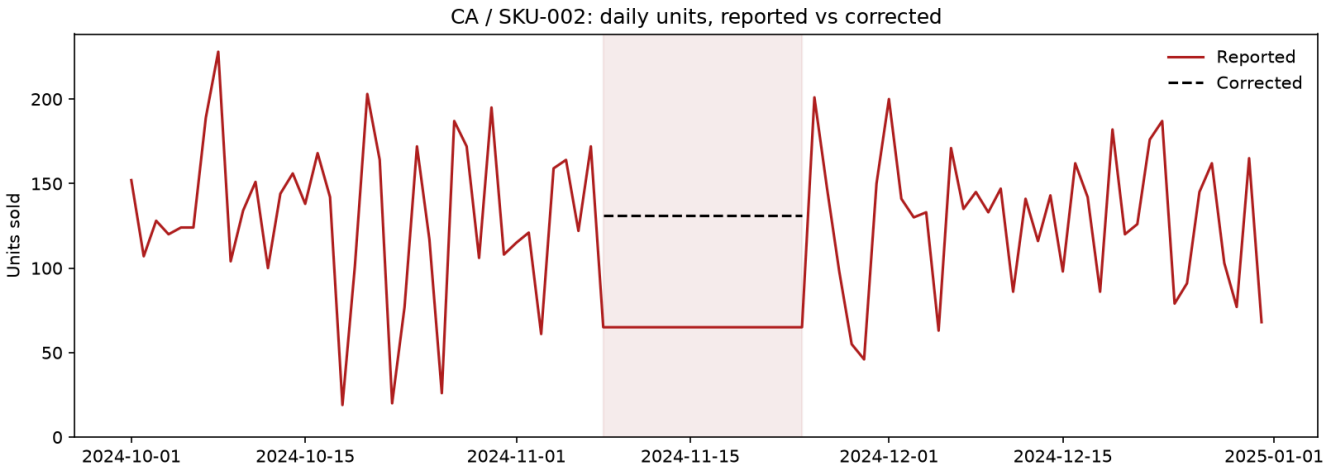
	SKU-002	SKU-004	SKU-012
National Q4 units understated	-4.6%	-4.7%	-4.8%
25 affected states, Nov 8 to Nov 24	about -50%	about -50%	about -50%
25 affected states, full Q4	about -9.5%	about -9.5%	about -9.5%

- National rank: SKU-002 is really #2 (reported #4); SKU-004 is really #3 (reported #5).
- Affected products, October to November: -19.8% as reported, -6.2% actual. All other products: -6.0%.
- Total units missing: 55,488 (under 1% of all Q4 units).

C. Charts



The 3 products in the 25 affected states (red) track the other 25 states (grey) until Nov 8, then drop to a flat line at half volume until Nov 24. The dashed line is the correction.



One example series: California, SKU-002. Normal daily swings, then a flat line at half height, then back to normal.

D. What looked off but wasn't

Our first lead was the wrong one. Daily totals drop about 44% on Nov 28–29, Dec 24–25 and Dec 31, and at first glance that looked like missing data. Broken down by state and product, every state and every product drops on the same days by a similar amount. Those are Thanksgiving/Black Friday, Christmas and New Year's Eve, so this is normal seasonality and needs no fix.

- **Short repeats in slow sellers:** products selling about 7 units a day sometimes repeat a number for up to 11 days by chance. Busy products never repeat for more than 5 days, which is why the frozen-value check uses a longer threshold for slow sellers.
- **Zero-sale days:** 1,386 records, all in slow sellers, where a day without a sale is expected.
- **One isolated spike:** MT / SKU-038 on Oct 2 (17 units). A single record, with no effect on totals or rankings.

E. How to reproduce

Requires Python 3.11+. From the repo root:

```
python -m venv .venv
source .venv/bin/activate      # Windows: .venv\Scripts\activate
pip install -r requirements.txt
python analysis.py
```

The corrected file is written to `output/sales_fixed.csv` and the charts to `charts/`.